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Keyboard Design and Operation: A Review of the Major Issues

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A search of the psychological, technical, and promotional literature was conducted to compile information relevant to key, keyboard, and operator characteristics. The most recent and significant articles were discussed and evaluated. Where possible, general conclusions have been drawn to aid the keyboard designer.

INTRODUCTION

Within the past decade, the proliferation of computers and office equipment has broadened interest in keyboards, typewriters, and related data entry devices. This paper reviews much of the applied research on the operator-keyboard interface. The major variables affecting keying performance are characteristics of the operator, the tasks he performs, the equipment with which he works, and his environment. A number of well-known human factors handbooks provide detailed data and design principles covering these variables (cf., Dreyfuss, 1959; McCormick, 1964). However, the impact of a number of additional variables such as the type of input materials used (alpha versus alphanumeric versus numeric), the performance measures employed (speed versus accuracy versus speed and accuracy), related keyboard characteristics (slope, key force-displacement), and subject input skill level remain unreconciled.

It is with these issues and their relation to characteristics of the operator that this review will be concerned. Hopefully, the results of such a review will serve as a basis for future design recommendations for keyboards which may in turn ultimately be incorporated into design standards.

OPERATOR CHARACTERISTICS

In conducting research on operator keying performance, special care must be taken to control or equate subject variables. Previous studies of manual data entry have identified a number of significant operator variables, including age (Leonard and Newman, 1965), fatigue (West, 1969), and level of training (Klemmer and Lockhead, 1962a).

Physiological Characteristics of Performance

The musculature of the hand places limitations on the ability of the fingers to respond in a keyboard task. Haaland, Wingert, and Olson (1963) measured the maximum pushing force for each finger of the adult male hand (Table 1). Haaland, *et al.*, attributed the differences in applicable force as reflecting the anatomical differences in the musculature of each finger. Also, it was observed that individuals whose fingers bent back when pushing perpendicular to a surface did not apply as much pushing force as those whose fingers remained rigid.

The data on maximum finger tapping rate

TABLE 1

Maximum Finger-Pushing Force (lb.)

	Finger				
	Thumb	*Index*	*Middle*	*Ring*	*Little*
Average	37	24	22	18	12
Range	30-43	17-31	17-26	12-22	6-19

(Haaland *et al.*, 1963).

correlates with these data on maximum pushing force. Early studies by Dvorak, Merrick, Dealey, and Ford (1936) indicated that the maximum tapping rate decreases from the index to the little finger (Table 2). The slight superiority of rates obtained with right-hand fingers may reflect their more frequent use. The rates for the four fingers of each hand decrease in order from index finger to little finger. Fox and Stansfield (1964) collected data on typists tapping with one or two hands. The modal tapping rate for alternate-hand tapping was once per 0.11 sec.; for same-hand tapping, it was once per 0.13 sec. Performance data indicate that for short periods, experienced operators may achieve rates of five or more strokes per second (Klemmer, 1971). This is in contrast to less frequent keyboard users for whom speeds on the order of one and one-half strokes per second have been observed (Smith, 1967).

In light of the data on finger strength, one would expect that fatigue would vary as a function of the finger being used. Haaland (1962) verified that the thumb is the most resistant to fatigue and that susceptability to fatigue increases as one moves progressively from the index finger to the little finger. The angle at which force was applied to the keyboard was also a factor, lateral force resulting in more rapid fatigue than downward force.

The thumb-index finger relationship has been studied by Binkhorst and Carlsöö (1962), Garrett (1971), Haaland (1962), Haaland, *et al.* (1963), and Lundervold (1951), among others. Binkhorst and Carlsöö used the technique of electromyography to study the musculature involved in thumb-index finger coordination. Their data is of interest with regard to the type of binary keyboard on aircraft control sticks.

Haaland, *et al.* (1963) observed that the thumb and fingers tire with sustained contractions. Practice with these muscles, however, can increase their capacity for contraction without fatigue. This latter finding is of some importance since measurements of actual operator performance indicate that experienced persons may make 56,000 to 83,000 keystrokes per day (Klemmer and Lockhead, 1962a).

The thumb is used primarily to provide "spacings" in current keyboard applications due to its limited flexibility. Also, positions at the extreme ranges of the thumb's motion are to be avoided in keyboard design, since such positions are possible only by a maximal contraction of the principal muscles involved.

Operator Capabilities

Human operators have definite limitations on their ability to accept and process data. These limitations often have direct relevance for the design of keyboard equipment and operator tasks. A theoretical review of factors which affect operator sensory-motor performance has been provided by Welford (1960). Another review, more directly concerned with the keyboard design aspects of human information processing, is contained in Hillix and Coburn (1961). As a guide to operator performance limits, Seibel (1964) reports information

TABLE 2

Maximum Tapping Rate by Hand and Fingers (Taps per Second Over 15-Sec. Test Interval)

	Hand							
	Left				*Right*			
	Little	*Ring*	*Middle*	*Index*	*Index*	*Middle*	*Ring*	*Little*
Tapping Rate	48	57	63	66	70	69	62	56

(Dvorak, *et al.*, 1936).

transmission rates for experienced typists as high as 17 bits per second.

A vast amount of research has been done on the input or perceptual processing aspects of operator performance. The variables which have proved important include the number of stimulus alternatives, their statistical structure, the number of signal sources per sense modality, the number of sense modalities, the temporal structure of incoming signals, and the coding format used. Poulton and Brown (1968), for example, found that teleprinter output was comprehended more rapidly when both capital and lowercase letters were used, rather than all capitals. Klemmer (1971) explored various groupings of digits for manual data entry. No effect on operator error rate was found, but speed was best with groupings of three or four digits.

Welford (1960) discussed central limits on information processing rates in some detail. The amount of recoding required by the operator in going from stimulus to response is a significant factor. In some cases, there is a one-to-one input-output relationship; in other cases it is several-to-one; and in still others, one input may require a sequence or combination of outputs, such as in multiple-press keying to be discussed later. Stimulus-and-response set compatibility is another well-documented factor. Furthermore, within limits, persons can vary the trade-off between speed and accuracy to give emphasis to one or the other (Howell and Kreidler, 1963).

The type of source material can limit operator performance. Shaffer and Hardwick (1969a), using both qualified and trainee typists, studied the effects of word lengths and word types (real words, nonsense words, and random consonants), on errors and error-detection probabilities. Each subject was instructed to type prose text as fast and as accurately as possible. If she detected an error, she was to retype the word before continuing. As expected, trainee typists made significantly more errors than qualified typists for all word types and lengths. Furthermore, across all test conditions, qualified typists were able to detect more errors than trainee typists. Yet for both groups errors increased and the probability of error detection decreased with increasing word length. A subsequent study by Shaffer and Hardwick (1969b) pursued this relationship between word length and word type. Response time per character was used as the dependent variable, since error rate was less than 1% for all conditions. Eight qualified typists were instructed, as before, to type as fast and accurately as possible. It was found that as the structure of the source material departed from real-word structure, operator response times increased significantly.

Requirements for short-term memory have been found to be a significant factor in keyboard tasks. Conrad (1966) found that errors in the use of a keyboard layout characterized by random-letter placement could be attributed to short-term memory breakdown rather than faulty response aiming. Additionally, Bowen and Guinness (1965) reported results of preliminary experiments indicating that keyboard design evaluations should be related to whether the task involves working from memorized codes or encoding directly from an input feature.

Characteristics of the response set imposed by keyboard task designs may also limit operator output. Significant variables include the number of alternative responses, the number of response channels, the complexity of the response monitoring requirements, temporal response rates, and response-response compatibility. These variables will be discussed subsequently as they relate to specific key-pressing tasks.

Measurement Techniques

A factor which influences the meaning of reported key-pressing speed and error rates is the metric used to measure operator performance. For some tasks, speed may be the most

valid measure of performance. Lincoln and Konz (1966) evaluated alternative configurations for a switch matrix and found that error rates were negligible in relation to speed differences across the different configurations. In other tasks, errors, although very infrequent, may be the more valid performance measure. Klemmer and Lockhead (1962a) found that average error rates for card-punch and bank-proof machine operators in different installations ranged between 0.02 and 0.06% (for undetected errors). They found that error rate differences between good and poor operators were much greater than speed differences, error rates being six to ten times greater for the poorest operators.

There are two frequently used approaches to the combined measurement of speed and accuracy. The most common of these is a "throughput" measure, of which typing tests are the best example, in which an arbitrary correction is made for each error. Performance is expressed in "words per minute" (wpm) or "net words per minute" (nwpm) after the adjustment of strokes subtracted for errors has been made. The second approach to measuring speed and accuracy simultaneously is in terms of an information metric of bits per second. In this approach, the correction for error is based on more rational grounds but is usually more difficult to apply.

A number of research studies have examined the relationship between operator speed and error rates. In general, these studies indicate that within subjects low speed and high error rates tend to go together (cf., Carlson, 1963; Conrad, 1966). Under some circumstances, however, a reverse relationship has been found (cf., Bergos, 1960). The latter finding appears to depend upon a speed-versus-accuracy tradeoff made by the operator. Howell and Kreidler (1963) found that performance on a key-pressing task varied with the emphasis of instructions (speed, accuracy, or speed plus accuracy). A conclusion from these studies is that when not provided a directional set it appears that people tend to set high standards for accuracy as opposed to such standards for speed. This is important for applications such as computer data entry when accuracy is of paramount concern.

Predicting Typing Proficiency

Martin (1956) reviewed the literature on aptitude tests for typing and provided an interesting historical perspective of this area. His general conclusion was that no well-validated typing aptitude test existed at that time. However, he did indicate that a task as near as possible to that required in the criterion situation might have some predictive value. To meet this need Martin developed a prototype test which appeared to have potential, but was not cross-validated.

Flanagan, Fivars, and Tuska (1959) conducted a development and validation study to show that a finger-tapping test could be used to screen typing applicants. The test was based on two important job elements: (1) the ability to tap with one finger at a time by controlling each finger separately and independently; and (2) learning to respond with a particular finger on recognizing a letter, number, or symbol. The results indicated that this test could serve effectively to screen applicants, especially where facilities were limited, equipment unfamiliar, or individual background and experience are diverse.

Droege and Hill (1961) proposed the development of conversion tables that would enable prediction of performance on an electric typewriter from performance on a manual typewriter. They collected performance data on 575 experienced typists typing on both manual and electric typewriters. The results based on typing test performance indicated that there was a substantial relationship between word-per-minute scores on both machines, but a lower relationship between error scores. On the average, individuals in the sample typed about 9 wpm faster and made about two fewer errors

on the electric typewriter than on the manual typewriter. It was concluded that due to the low correlation between error scores on the two machines, manual typewriter performance does not provide a satisfactory basis for predicting electric typewriter performance.

More recently, West and Bolanovich (1963) developed parallel forms of an office typing proficiency test. These tests were used as employment screening tests and for assessing the results of industrial training programs. Reliabilities of 0.77 for speed and 0.88 for errors were found for a production test criterion (business letters, tables, rough drafts), and 0.97 and 0.73 for straight copy work. These values are based on the performance of 168 advanced typists. Validity coefficients ranged between 0.56 and 0.80, with dichotomized rankings by two instructors serving as the external criterion.

Leonard and Carpenter (1964), recognizing the need for a reliable predictor of typing skill, attempted to evaluate the utility of a five-choice key pressing task. Based on a sample of 92 female students with various keyboard experience, they found that performance on a serial choice-reaction test showed a significant and appreciable correlation with measures of typewriting achievement over a period of nine months. Thus, there are several test tasks which may be used to predict speed typing proficiency. However, when the criterion task includes a measure of accuracy the predictive validity of these tests decrease.

Acquisition of Key-Pressing Skill

West (1957) reviewed literature concerning instructional procedures, materials, and equipment from the earliest known studies to mid-1956. Aspects of technique, speed, accuracy, motivational factors, and standards of achievement were discussed. Among his conclusions for the development of an effective key-pressing training program are the following: (1) Use only real words and sentences for source material, never random or nonsense combinations. The studies by Shaffer and Hardwick (1969a, b) substantiate this conclusion. (2) Stress key-pressing speed early in training, although, not to the point of muscle fatigue. (3) Use lenient error standards during initial training. (4) Do not impose a set pace on all students. Allow each one to proceed at her own rate. (5) Permit the student to have complete sight of the keyboard. This maximizes the acquisition of keyboard character location and good stroking technique.

As can be seen from the above, the role of feedback is important to the issue of key-pressing training. It has been hypothesized that there is a transition from visual feedback dependence to a reliance on kinesthetic or proprioceptive feedback in the acquisition of a complex pyschomotor skill (Fleishman and Rich, 1963). West (1967) applied this line of research to the acquisition of typewriting skill. His results showed an increase in kinesthetic dependence from the beginning to the intermediate levels of proficiency and then no change through the expert level. Performance with kinesthetic feedback alone was always poorer than with "all-senses" feedback. Deprivation of vision had no effect on speed but it did produce large increases in errors. This finding suggests the desirability of the unlimited use of vision in learning to type, since visual feedback appears to act as a source of guidance for responses, as a source of positive and negative feedback, and as a means for reducing the tensions and anxieties associated with nonvisual work (West, 1957).

Augmented, supplementary, or extra feedback refers to any feedback added to a standard task, i.e., any feedback over and above kinesthetic feedback. Leonard and Conrad (1963), using postmen as subjects, taught them a logical association between pairs of keys on a ten-key keyboard and the letters of the alphabet. With speed and accuracy stressed equally, it was shown that subjects were able to improve steadily in speed without loss of accuracy

without receiving any augmented feedback. These data lend support to the hypothesis that augmented feedback can be withdrawn as soon as keyboard vocabulary has been learned well enough for the subjects to produce the items without errors. Once the augmented feedback is withdrawn, it is assumed that subjects are able to provide sufficient intrinsic feedback to maintain high accuracy. This assumption is consistent with the results of studies of experienced typists (Diehl and Seibel, 1962; Galitz, 1965).

Leonard and Newman (1965) studied performance with a task in which subjects were required to transfer information rapidly from different modality displays to five-position keyboards. The subjects, postmen or naval volunteers, were able to achieve and maintain high accuracy under a range of input modality codes. In an auditory display condition, immediate feedback was necessary to maintain high performance. The authors concluded that there are two possibly overlapping phases in the acquisition of a new sensory-motor skill. During the initial phase, the subject must learn a vocabulary, establish stimulus response relationships, and master the keyboard. During the second phase, improvement in performance takes place. Leonard and Newman suggest that augmented feedback and providing subjects with an "accuracy" set are important only during the first phase. During the second phase, it is assumed that augmentation is not essential because the subject provides his own feedback. Furthermore, the authors suggest that the first phase may be accomplished very quickly; it ends when subjects know what to do, how to do it, and have sufficient time to make each response.

This finding is somewhat at variance with the recommendations of West (1957), cited earlier, on a speed-versus-accuracy emphasis during early training. The keyboard size and character set used by Leonard and Newman is much smaller than the normal typewriter, which might account for their discrepant results.

Leonard and Carpenter (1964), however, found that performance using a numerical display and five-finger keyboard correlated highly with subsequent typewriting achievement. Thus the evidence is equivocal on whether to emphasize speed, accuracy, or both during training.

INDIVIDUAL KEY CHARACTERISTICS

Design Standards

No single industry or military standard for basic key characteristics, such as size, operating force, or displacement, has been formally established. Pollock and Gildner (1963), in a review of manual computer input devices, summarized the characteristics of commercially available keyboards. According to their report, considerable variation existed in keyshape (square, round, rectangular), operating force (ranging from 2 to 8 oz., or 57 to 227 g.) and displacement (0.067 to 0.625 in., or 0.71 to 1.59 cm.). Harkins (1965) also acknowledged the existence of a wide variety of key configurations. It is due to design conventions rather than empirical data, that keys of 0.50-in. (1.27-cm.) width or diameter are found on most typewriter and ten-key keyboards today. The typical spacing between key centers on these keyboards is 0.75 in. (1.81 cm.). Some ten-key adding machines have the same key size but reduce distance between key centers to 0.69 in. (1.75 cm.).

Dreyfuss (1959) recommended as a human factors standard that operating forces for keys in a keyset range between 4.1 and 11.0 oz. (117 and 312 g.) with 0.187 in. (0.47 cm.) displacement. In addition, he recommended a maximum key size of 0.50-in. (1.27-cm.) width and 0.438-in. (1.11-cm.) length. The United States MIL-STD-803 A-1 (1964) lists 32 oz. (908 g.) as the optimum operating force for an individual pushbutton, a value too high for effective keyboard operation. In the face of these rather broad recommendations, keyboard manufac-

turers have generally selected their own design criteria for individual keys.

Key Parameters

Considering the many possible combinations of key size, shape, force, and displacement, it is surprising that so few factorial experiments have been conducted to determine their effects on operator performance.

Deininger (1960) conducted the best-known and perhaps the most inclusive research on the effects which such characteristics have on performance. For the four main variables of key design studied—size, force, displacement, and feedback—he noted only small differences between operators. The subjects' task was to key in "telephone numbers" with different ten-key pushbutton sets. Deininger found that the largest difference in performance resulted from varying the button-top size. Increasing the size of a square button from 0.375 to 0.50 in. (0.95 to 1.74 cm.) reduced keying times from 6.35 to 5.83 sec., while it reduced errors from 7.3 to 1.3%. Further, Deininger found that varying the force from 3.5 to 14.1 oz., (100 to 400 g.) or varying the maximum displacement from 0.03 to 0.19 in. (0.08 to 0.48 cm.) produced insignificant differences in subject performance. Subjects' reports indicated a preference, in general, for the light-touch keys and a definite dislike for keys requiring greater displacement. Deininger's results must be interpreted somewhat cautiously for three reasons: (1) the results may be considered applicable only to ten-key numeric keyboards rather than to a full alphanumeric keyboard; (2) the reported findings probably included learning effects since there was no indication that asymptotic performance was attained; and (3) changes in button-top size were accompanied by changes in the size of the character on the key, thus confounding the effects of the two variables.

Kinkead and Gonzalez (1969), using eight experienced typists as subjects, evaluated the influences of key force and displacement on key-pressing performance. They found that throughput was best when both force and displacement were at relatively low levels. They recommend as optimal operating point key forces of between 0.9 and 5.3 oz. (25.5 to 150.3 g.) and total key displacements between 0.05 and 0.25 in. (0.13 to 0.64 cm.).

In general, the results of the few studies of individual key force and displacement are that these parameters (within certain limits) have little effect on the keying performance of experienced operators.

Key Feedback

In normal typing and other keypressing tasks, there is kinesthetic feedback from depressing the key, auditory feedback from the key depression and/or activation of the print mechanism, and visual feedback from the keyboard and output display.

Generalizing from the psychological literature on learning, it might be reasonable to expect that feedback is necessary for efficient keying performance. However, review of the literature on keying performance indicates that this may not be the case. Deininger (1960) found no statistical differences in speed and error performance when auditory and kinesthetic feedback was added to pushbuttons having 0.125-in. (0.32-cm.) travel and 3.5-oz. (99.2-g.) force. Diehl and Seibel (1962), using 16 experienced secretaries, stenographers, and typists as subjects and four key feedback conditions (normal, visual masking, auditory masking, and both visual and auditory masking), found no significant performance differences when measuring net words per minute on a speed typing test. Galitz (1965) found that four of twelve experienced typists stated that the absence of auditory feedback, which normally results from key depression, "bothered" them. However, these opinions were not compared with typing performance.

Kinkead and Gonzalez (1969) found that

the absence of kinesthetic snap-action feedback did not affect typing speed. In fact in this experiment operators using a snap-action keyboard made significantly more errors. These authors concluded that this type of kinesthetic feedback is not a desirable key parameter.

Other key feedback features, such as bottoming, breakout force, etc., may influence operator keypressing. However, these characteristics have not been studied to any large degree.

KEYBOARD CHARACTERISTICS

Design Standards

The character arrangement of the alphanumeric keyboard dubbed "QWERTY" has evolved into the American standard typewriter layout (Phillips and Kincaid, 1971). This fact was recently formalized by the subcommittee of the U.S.A. Standards Institute, Sectional Committee X4, Office Machine Group (Lohse, 1968a, b). The reasons given for this action were the costs and time which would be involved in retraining the millions of existing QWERTY operators. In addition, the QWERTY keyboard has been used as the point of departure for the design of the ASCII (Lohse, 1968a) and the USASCSOCR Dual Case (Lohse, 1968b) keyboards. An example of the strength of tradition associated with the QWERTY keyboard was the mandatory retention of the semicolon key as a "home" character on the ASCII keyboard. This recommendation was made by a U.S. Standards task group, despite the documented lower incidence of semicolon use relative to the colon (Avanzino, Garland, Horne, Richardson, Stork, and Umstead, 1969, Article E6. 1).

It is for reasons such as the above that Remington and Rogers (1969) suggested that human factors research on performance as a function of a keyboard layout "...would be only of academic interest" (p. 4). Instead, they recommended that research be devoted to those areas where existing criteria for standards were insufficient: numeric key layouts, optimum key displacement, interlocks and tactile feedback. Dolotta and Selfridge,[1] in their recommendations for a computer time-sharing terminal, stressed that such a terminal should have the characteristics of an electric typewriter. The recommendations made in that report were based on questionnaire responses by users and potential users of time-sharing computers. Although Dolotta and Selfridge indicated that user needs should take precedence over tradition, their recommended keyboard is the basic QWERTY arrangement.

It seems safe, therefore, to conclude that the QWERTY keyboard is the de facto standard layout for communications and computer interface keyboards. This has occurred despite suggestions that the arrangement is less than efficient (Phillips and Kincaid, 1971).

Key Grouping and Key Performance

Barmack and Sinaiko (1966), in their comprehensive review of the man-computer interface, enumerated the following devices as suitable for the input of alphanumeric information: standard typewriter keyboards; experimental keyboards; multiple-press (chord) keyboards; cathode-ray-tube (CRT) displayed lists and lightpens; and character- and speech-recognition systems. These areas will be discussed in turn, with the exception of the last two nonkeyboard data-entry methods.

Standard typewriter keyboards. As indicated in the preceding section, the QWERTY keyboard arrangement has become the accepted standard. An alternative arrangement, the Dvorak Simplified Keyboard, was developed in 1936 (Dvorak, *et al.*) The keys on this board are arranged according to frequency of letter occurrence in common English usage, with the

[1] Dolotta, T. A. and Selfridge O. G. Personal communication, September 1968.

most commonly used letters on the home row. Strong (1956) conducted a U.S.-government-sponsored study in which he compared performance on the Dvorak Simplified Keyboard with the standard QWERTY arrangement. Ten subjects (all QWERTY typists of varying experience) were retrained on the Dvorak keyboard until they regained their previous speed as measured in gross words per minute. Retraining took an average of 28.1 days. This retrained group and a comparable group, using the QWERTY keyboard, received four hours of daily instruction on their respective keyboards. At the end of the experiment, the QWERTY group was superior in both speed and accuracy on a 5-min. typing test. No performance differences were found between the groups on a 1-min. test. Strong concluded that there was no justification, based on these results, for the adoption of the Dvorak keyboard for government use.

There has been a recent resurgence of interest in the Dvorak Simplified Keyboard. It has been proposed that this keyboard is superior in terms of ease of learning, likelihood of error, fatigue rating, and speed of entry; therefore, serious consideration should be given to the adoption of this or some similarly designed keyboard (Dunn, 1971). The Dvorak keyboard has been used with impressive results in several prototype programs, to train children and adult computer programmers (McCauley and Parkinson, 1971). The ultimate goal of these programs is to "...introduce the Dvorak Simplified Keyboard into the noncommercial student market and into the data processing industry as a data entry device" (McCauley and Parkinson, 1971 p. 32).

Another possible keyboard arrangement is an alphabetical sequence (A to Z) such as that proposed for use by stockmarket brokers (Avakian, 1968). Barmack and Sinaiko (1966) and Klemmer (1971) both state in their reviews that this arrangement does not appear to be satisfactory as a standard for normal keyset use. Hirsch (1970) compared the performance of nontypists on both the QWERTY keyboard and an alphabetically arranged keyboard. The subjects typed familiar material, such as their names, addresses, etc., with instructions which tended to stress accuracy over speed. Following seven hours of practice, those subjects who trained on the alphabetically arranged keyboard were at the same level of proficiency on this keyboard as they had been on the pretest using the QWERTY keyboard. Those subjects who practiced with the QWERTY board had achieved significant improvement. An additional study (Michaels, 1971) used skilled, semiskilled, and unskilled typists as subjects to compare performance on an alphabetically arranged keyboard with that on a QWERTY keyboard. His findings were consistent with those of Hirsch (1970) in that no advantages were found for the alphabetic arrangement. Regardless of initial skill level, operators were as fast or faster using the QWERTY arrangement.

Experimental keyboards. Keyboards of unique design have been proposed for both general and special-purpose uses. For example, Rosberger (1965) proposed a binary keyboard in which each key could have a double function. This was accomplished, without use of a shift key, by making use of the intersection between keys. The keyboard was proposed for one-handed touch typing. Camuso (1968) reported on an "overlay keyboard," to be used with a cathode ray tube for computer input. Lewin (1967) proposed a stylus-operated "sequential keyset" to input ASCII codes. However, there are no known reports in the literature regarding comparison of these particular keyboards with standard keyboards in terms of speed and accuracy of operation.

Some experimentation has been conducted, however, to develop keyboard design principles and to evaluate alternative configurations. Hillix and Coburn (1961) reviewed the literature on keyset design as it related to real-time computer data entry. They recommended that keysets should be evaluated in an operational

context, since the speed and error properties of a keyset are ultimately under operator control. In line with their recommendations for consideration of a wider range of input device types, the Naval Electronics Laboratory conducted a series of keyset investigations (Hufford and Coburn, 1961; Newman, Coburn, and McMahan, 1962).

Newman, *et al.*, (1962) compared binary-coded keys to pushbuttons for efficiency of computer data entry. They trained subjects to time and error asymptotes on each experimental device. Performance was found to be best on a four-key binary keyset. In addition, they found that experienced operators could perform simple stimulus encoding tasks during data entry, thus facilitating speed of performance. A critical factor was that the speed and error trade-offs made by the operators were affected by keyset design.

Davidson (1966) has suggested adding two control keys to the pushbutton telephone and using it as an alphanumeric computer input device. These two additional buttons would be used to identify which character on a data button was being specified. For example, the right control key would indicate that the right character on the data button was the command.

Kramer (1970) has conducted both field and laboratory studies to determine what problems, if any, were associated with using the pushbutton telephone for data entry. In one study he compared the arrangement proposed by Davidson above with an arrangement where the number of presses of a pushbutton indicated which character position was specified. The control key arrangement in both of its modes (position code first or position code last) was superior to the other in terms of user accuracy, although both methods resulted in high error rates. There were enough residual questions unanswered both from an individual operator and systems viewpoint that definitive recommendations could not be made.

Smith and Goodwin (1971) made a survey to determine the ambiguity which would result from using the touch-tone telephone pad for alphabetic entry of proper names via single-stroke keying (e.g., the letters *A*, *B*, or *C* would be entered by pushing the "2" button). This might eliminate the high error rates associated with double stroke keying found by Kramer (1970). In two separate samples of American surnames only 7 out of 256 and 8 out of 345 names, respectively, resulted in exact numeric duplication. Further, it was suggested that the context associated with the entry of the data could be used by computer software to resolve any ambiguities.

In a study for the Federal Aviation Agency, Paul, Sarlanis, and Buckley (1965) compared two experimental 16-key keyboards designed for one-hand operation. The keyboards differed in that one keyboard had the numerics arranged in the adding-machine format (Figure 1c), while the other had them in the touch-telephone format (Figure 1a). The subjects were air controllers and were matched for group assignment on the basis of a typing pretest. The stimulus materials were random numbers, random letters, valid aircraft identifiers, and random aircraft identifiers (i.e., two random letters and three random numbers). Performance on the top-to-bottom keyboard was superior both for alphanumerics and for letters alone. However, no differences were found in keying numerics. This result might be explained by the relatively common use of the two numeric layouts tested.

Due to the lack of standards for the layout of numeric keyboards, several arrangements are being used, the most common of these being the adding machine and the touch-telephone layouts. Conrad (1966) investigated the effect of compatibility of the stimulus-response of a numeric keyboard layout upon performance. He compared a high-compatibility keyboard (Figure 1a) with a low-compatibility one (Figure 1b) and predictably, performance was superior on the high-compatibility board. Conrad indicated the importance of standardizing numeric keyboards, since the observed

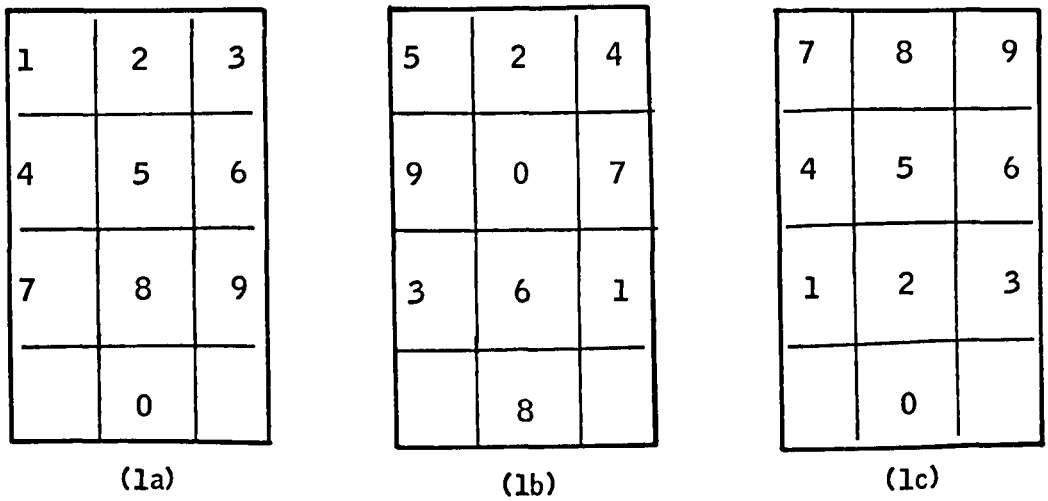

Figure 1. Various numeric keyboard arrangements.

errors he attributed to short-term memory rather than key pressing. A second study (Conrad and Hull, 1968) compared the high-compatibility telephone keyboard layout (Figure 1a) with an adding-machine format (Figure 1c). Using housewives as subjects and speed and accuracy as the metric of performance, the touch-telephone keyboard layout was found to be significantly superior.

Other methods of numeric data entry have also been studied. Minor and Revesman (1962) compared four numeric entry devices: a ten-key keyboard with full visual feedback, a matrix keyboard, levers, and rotary knobs. Using inexperienced production employees (the "target" population) as subjects, they found that fewest errors were made on the ten-key keyboard. Performance on this board was faster than for levers and knobs for data entry and equivalent to the matrix keyboard. The ten-key keyboard was also the most preferred. The subjects, however, were not at asymptote at the end of the study, thus making final performance generalizations impossible. Also, Seibel[2] has questioned whether these results would hold if the keyboard did not provide full visual feedback.

Hufford and Coburn (1961) compared "standard" size with miniaturized numeric keyboards as part of the Naval Electronics Laboratory computer-entry device program. Three miniature keysets were studied: "touch" keyset with the numbers arranged in a circle, a pencil-stylus probe keyset, and a finger-stylus probe keyset. Both stylus probe keysets were rectangular in shape. While performance on the standard size and the probe keysets was equivalent in terms of speed and accuracy, touch sets produced noticeably inferior performance. The authors hypothesized an eventual superiority for the miniature keysets due to their dependency on finger rather than gross muscle movements, although their subjects were reported to have been trained to asymptote.

Multiple-press keyboards. Multiple-press or "chord" keyboards, in which several keys must be depressed for a single character to be output, have been suggested as much faster than the QWERTY keyboard. Chording is the principle which underlies the Stenowriter stenotype machine (Galli, 1960) and the Word-Writing machine (Ayres, 1965). Performance claims of

[2] Seibel, R. Data entry devices and procedures. In H. P. VanCott (Ed.) *Human Engineering Guide to Equipment Design.* (2nd ed.) Washington: U.S. Government Printing Office, in press.

200-300 wpm have been made for these machines.

The basic premise of the multiple-press keyboard is that the decision time required to make two simultaneous presses with two hands is small relative to the time required to make a single multiple press with one hand. Further, the additional time required to learn operation of a chorded keyboard is repaid by the gain in net output. Support for this premise was supplied by Ratz and Ritchie (1961) who ranked each of the 31 possible one-hand chords in terms of choice reaction time. The Ratz and Ritchie keyboard had ten keys, one for each finger, and was used in conjunction with a compatible light display. These researchers concluded, with the qualification of high stimulus-response compatibility, that speed on a chord-keyboard was determined by motor constraints rather than decision time. They observed, in addition, rapid attainment of asymptotic performance. This latter finding, however, was questioned by Seibel (1962). Seibel replicated the Ratz and Ritchie study and found continued improvement in discriminative chorded reaction time over 30 days.

Pursuing this theme, Seibel (1964) conducted a feasibility study for a chorded typewriter device which required that operators learn a shorthand input code. Despite the requirement for learning an expanded vocabulary, rapid performance improvements and high typing rates were noted. Thus, Seibel indicated that despite the unanswered questions regarding chorded inputs, such methods could result in the goal of speeded data entry.

Cornog, Hockman, and Craig (1963) evaluated a double-binary chorded keyboard for use in a mail-sorting task. The results of this study indicated the feasibility of such a keyboard for this purpose. Bowen and Guinness (1965) conducted an experimental evaluation of three keyboards for semiautomatic mail sorting. One keyboard was a standard typewriter and the others were large and small chorded keyboards. The format of the large chorded keyboard corresponded to the 24-key double-binary keyboard used by Cornog, *et al.* The results, based on matched groups of six subjects each, show that for encoding tasks where the digitation for each address must be remembered, the large chorded keyboard is superior. Where information must be selected from an address (extraction encoding), the typewriter and large chorded board were equivalent. Yet for skilled operators, the study indicated the performance differences between different keyboards tended to be small. The critical questions, therefore, might well be directed to training and practice requirements.

Another study of keying on a post-office task was conducted by Conrad and Longman (1965). They trained two groups of postmen for seven weeks, one group on a chord keyboard, the other on a standard typewriter. The chord typists were initially faster, and over the seven weeks both groups improved at the same rate. Although the typewriter group was more accurate, it also received immediate error feedback whereas the chord group did not.

Therefore, there are data to support the hypothesis that keyboards which tend to minimize the number or the distance of finger-reaching movements are capable of the fastest operation, particularly for special-purpose tasks.

Keyboard Slope, Tilt, and Size

There is a relatively small literature on the question of typing performance as a function of keyboard inclination, including both slope and lateral tilt. Moneta (1960), in a review of the literature of fatigue as it relates to keyboard operation, reported that the British standard for manual typewriter keyboard slope was 30° from the horizontal. She indicated that this angle appears to be based on convention rather than empirical data, since it has been shown that at an angle of 30° it is difficult to hold the forearms without effort (Lundervold, 1951). Moneta concluded that some lower angle of

slope would be preferable, but did not indicate a particular value.

Scales and Chapanis (1954) studied keyboards which varied in slope from 0 to 40°. Sixteen subjects performing a task of keying ten-character alphanumerics showed no significant differences in either typing errors or speed as a function of slope. The operator preference data indicated that some slope was desirable, half of the preferences being between 15 and 25°. Dreyfuss (1959), in his tables of anthropometric data, indicated that an 11° slope was the optimum for a typewriter keyboard, with 20° the maximum permissible.

Galitz (1965) investigated the effects of the slope of a CRT-console keyboard on typing performance. The subjects were all experienced electric typewriter typists and were given no special training on the experimental keyboard prior to the three-day study between the slopes of 9, 21, and 33°. No significant performance differences were found.

Galitz did find that a 21° slope was preferred. This was the slope which most closely approximated the subjects' normal equipment (16 to 17°). He concluded that keyboards should permit a slope adjustment between 10 and 35° in accord with operator preference, for there appears to be little performance difference over this range. Several of the new electric typewriters (e.g., IBM Selectric) offer this option.

Key tapping has been found to be a valid predictor of typing performance (Flanagan, *et al.*, 1959). Therefore, the conclusions drawn by Creamer and Trumbo (1960) from a key-tapping task are relevant to the issue of lateral keyboard tilt. Using a keyboard shaped like an accordion, Creamer and Trumbo investigated the effects of tilts of 0, 22, 44, 66, and 88°. The subjects were required to tap keys in a particular sequence, alternating fingers and hands. The subjects made the maximum number of errors at a 0° slope. There was a slight monotonic decrease in errors from 22 to 88°. Creamer and Trumbo pointed out that the worst condition resembles the tilt of a typewriter keyboard. However, if one were to compare respective typing surfaces, a tilt condition of 22° is closer to the typewriter norm than is 0°.

Two additional studies which present evidence regarding the relationship between performance and tilt were reported by Kroemer (1964, 1965). Kroemer indicated that keyboard tilts of 15° proved significantly slower (raw strokes/second) and more fatiguing (subject termination of trial) than an experimental "K-keyboard." The "K-keyboard" differs substantially from the "normal" alphanumeric typewriter arrangement in that it has separate keyboards for the right and left hands with the keys arranged in arched rows. Four angular tilts were investigated (0, 30, 60, and 90°). Kroemer reported that an inclination of 30 or 60° produced significantly faster and less fatiguing stroking performance than either 0 or 90° inclination, a finding which is in essential agreement with that reported by Creamer and Trumbo (1960).

Even fewer data have been published on the effects of keyboard size. As noted earlier, Hufford and Coburn (1961) compared three miniature keysets with a standard-size, ten-key numeric keyboard (3. 06 X 2.125 in. or 7.78 X 5.40 cm.). The two probe-actuated miniaturized keysets measured 0.97 X .69 in. (2.46 X 1.75 cm.). Yet, no differences were observed in terms of errors between the probe-actuated miniature and the standard-size boards. A "touch"-operated miniature keyset (1.125 in., or 2.86 cm., outside diameter) produced significantly poorer performance than the others in terms of both speed and accuracy. However, the authors suggested that even this keyboard presented promise for certain task conditions.

The general conclusion which may be drawn from the above is that key-pressing performance by experienced typists appears to be remarkably stable over a wide range of keyboard slopes, tilts, sizes, and shapes.

Interlock Systems

Keyboard interlock systems are designed to prevent the outputs from two or more simultaneously depressed keys from either jamming the print mechanism or outputting an invalid key code. Kinkead and Gonzalez (1969) identify three types of interlock systems: mechanical, structural, and functional.

Mechanical interlock. In most electric typewriters there is no mechanical lockout. Rather, simple logic circuitry limits the rate of output from key depressions. Because print mechanisms normally have a maximum operational speed which is lower than operator keying capability, certain design features are employed to overcome the speed limit imposed by the printer and prevent jamming of the print mechanism. The three common types of structural interlocks are: two-key roll, one-character memory, and multiple-key depression.

Two-key roll incorporates an electronic circuit which limits the rate at which codes generated by key depression are fed to the print mechanisms or other output circuit. When the first key is depressed, it generates both a letter code and a blocking code. The blocking signal prevents transmission of a letter code from the depression of a second key. However, as soon as the first key is released, the code for the second key is transmitted and another blocking signal occurs. In this way, key-depression rate can momentarily exceed the speed limits of the print mechanism or other machine features.

One-character memory is found in the IBM Selectric typewriter. Codes generated by key depressions are routed through a one-character automatic-erase, storage circuit. A code for the last character depressed is always in storage. Thus, it is possible to exceed the print mechanism capability if the series of key depressions does not exceed a factor of two. An excellent description of the two-key roll and newer *N*-key roll electronics has recently been published by Kallage (1972).

Multiple-key depression is found in some computer terminals. It is identical to one-character memory, except that the characters are not stored.

Functional interlock. This system involves no mechanical or electric mechanism to prevent simultaneous depression of two or more keys. If the operator depresses two keys simultaneously there results the simultaneous activation of two letter strike bars or the transmission of two letter codes. However, when this occurs in an electronically linked system these letter codes combine and produce either a new valid code (if the summation gives a different but recognizable code or chord) or gibberish which is unrecognizable by the printer mechanism.

A number of studies have investigated the influence of keyboard interlock system characteristics on operator performance. Minor (1964) investigated the effects on operator speed and accuracy of structural interlocks requiring 125- and 77-msec intercharacter depression delays. Using a special-purpose IBM Model A design typewriter, the subjects, all experienced typists, typed address labels under instructions to be as accurate as possible. When the subjects detected that they had made an error, they were to retype the entire line in which the error occurred. Six 15-min. test trials were given for each interlock condition. A six-day period elapsed between interlock condition sessions. Minor found that undetected operator keying error rate was significantly less for the 77-msec delay (8 strokes-per-second maximum keying rate) than for the 125-msec delay (13 strokes-per-second maximum keying rate). Although both the 77- and 125-msec minimum intercharacter times were above the 50 msec hypothesized by Minor as the limit for skilled typists, his subjects reported that they infrequently "ran" the keyboard into the interlock during test sessions. He concluded that the influence of different interlocks will vary as a function of key force displacement, input material, and operator skill level.

Based on a series of UNIVAC-sponsored studies which explored typing speed as a

function of keyboard characteristics, Galitz (1965) recommended that key interlocks to minimize errors at high input rates should disengage at three-quarters of the downstroke distance. The interlock is activated for the remaining quarter of the downstroke and the first quarter of the upstroke. Later, Galitz (1966a) compared speed and error performance of typists using keyboards with momentary key interlocks to that achieved on machines with interlocks such as the above. The findings of this study indicated that both speed in net words per minute increased (45.11 to 64.33) and raw errors per minute decreased (4.83 to 1.81 per minute) when subjects used the momentary interlock keyboard. The proportion of errors of omission alone dropped from 90% with the long-interlock keyboard to 17% with the momentary-interlock keyboard. Based on these findings Galitz recommended that key interlocks should be momentary in nature, engaging and disengaging on the down keystroke at a point between 65-75% of the downward key travel.

In a subsequent study Galitz (1966b) evaluated the momentary-interlock keyswitch developed by MICROSWITCH. Again he found a clear superiority for the single-pulse, momentary-interlock switch over the longer interlock switch which he had previously evaluated. This difference was observed for both speed and error measures.

Daniels and Graf (1970) reported results of a study designed to assess the maximum typing speeds obtainable with various keyboard interlock systems. Using a MICROSWITCH Solid State Key dual-mode keyboard linked to a computer, these investigators measured intercharacter keying times as low as 4 msec for the digram "he." This occurred with the two-key-roll interlock configuration. The authors concluded that if keyboard interlock systems are not to inhibit high-speed typing performance, they must be capable of accurately handling data input rates in areas of 100 gross words per minute.

Operator Preference

Frequently, in the studies reviewed, questionnaires were distributed to the subjects to determine operator preferences for various key parameters. Lee and Snodgrass (1958) reported, however, no significant relationship between initial keyboard preference and subsequent keying performance. Further, they found little relationship between preference before and after using the experimental keysets. Kinkead and Gonzalez (1969) also found low correlations between preference and performance, except in the case of a keyboard with high force and low displacement. Each of the experienced typists complained that this keyboard was difficult to type on and all performed accordingly. Kinkead and Gonzalez further found that their subjects were inconsistent in their preference ratings.

Anecdotal evidence suggests that typists frequently report that they prefer the machine they are accustomed to using, but will change their preference after using a new machine for some period of time. Therefore, one may conclude that preference ratings must be interpreted very cautiously and should seldom, if ever, be the sole basis for design recommendations.

SUMMARY AND CONCLUSIONS

A review of the recent literature indicates that much research effort has been expended on the operator-keyboard interface. However, as the following statements indicate, there are few definitive findings on which to base design standards. The following are general design considerations as abstracted from the discussed reports:

Key dimension. Key diameters of 0.5 in. (1.27 cm.) and 0.75 in. (1.81 cm.) center-to-center spacings are most typical.

Force and displacement. Ranges of 0.9 to 5.3 oz. (25.5 to 150.3 g.) force and 0.05 to

0.25 in. (0.13 to 0.64 cm.) displacement are preferred by operators, although force and displacement have slight effect on keying performance.

Feedback. Visual, auditory, or kinesthetic feedback shows little effect on performance by experienced operators, but visual feedback appears to be important during training.

Keyboard size. Miniaturized keyboards used for touch operation show performance decrements, but size itself has little effect on self-paced or sequential operation.

Keyboard inclination. Ten to 35° seems an acceptable range for inclination of the working surface. However, this has questionable influence on operator speed and accuracy, even though most typists preferred some keyboard inclination.

Numeric cluster. A top-to-bottom, left-to-right cluster (touch-telephone format) seems to be the superior arrangement from the available evidence.

Operator preference. Operator preferences do not correlate highly with measured performance.

Entry method. Chorded entry may be faster than sequential entry, especially for some special tasks, such as mail sorting.

Although research on keyboard design and use continues, the sponsorship and motivation for such work often precludes the systematic study of any single keyboard parameter. The majority of the current research findings in the open literature have been undertaken to evaluate specific product lines. It is very seldom possible, therefore, to obtain objective data on the relative speed or accuracy of performance across various generic types of keyboard designs. This situation results in a proliferation of keyboard and task-specific data. Unfortunately, such data are often incompletely reported or involve unique metrics, making interstudy comparisons difficult or impossible. Consequently, the major issues of today are the same as those of a decade or more ago. Furthermore, until there is a systematic effort to produce relating, unambiguous, quantitative data on the parametric relationships between keyboard design and operator performance, such issues will remain unanswered.

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REFERENCES

Avakian, E. A. Applications and utility of alphabetic clusters, the non-QWERTY keyboards. Paper presented at the Workshop on Keysets and Keyboards of the IEEE Man-Machine Systems Group, Gaithersburg, Maryland, December 1968.

Avanzino, L., Garland, S., Horne, B., Richardson, T., Stork, E., and Umstead, H. Proposed U.S.A. standard: Logically bit-paired ASCII keyboard arrangement. USA Standards Institute Subcommittee X4-A9, Task Group 1, April 1969.

Ayres, A. Word writing machine producing closed-up printing in response to simultaneous actuation of keys. Patent Specification 3,225,883, patented December 28, 1965.

Barmack, J. E. and Sinaiko, H. Human factors problems in computer-generated graphic displays. Institute for Defense Analyses SD-50, April 2, 1966. (AD636170)

Bergos, R. An analysis and comparison of typescript errors made on electric typewriters with those made on non-electric typewriters. Abstract in *The National Business Education Quarterly*, 1960, 29, (4), 9-10.

Binkhorst, R. A. and Carlsöö, S. The thumb-forefinger grip and the shape of handles of certain instruments: An electromyographic study of the muscle-load. *Ergonomics*, 1962, 5, 467-470.

Bowen, H. M. and Guinness, G. V. Preliminary experiments on keyboard design for semi-automatic mail sorting. *Journal of Applied Psychology*, 1965, 49, 194-198.

Camuso, A. (Ed.) New concept in man-machine interface devices. *Computer Design*, September 1968, 7, 32.

Carlson, G. Predicting clerical error in an EDP environment. *Datamation*, February 1963, 9, 34-36.

Conrad, R. Short-term memory factor in the design of data-entry keyboards. *Journal of Applied Psychology*, 1966, 50, 353-356.

Conrad, R. and Hull, A. J. The preferred layout for numerical data-entry keysets. *Ergonomics*, 1968, 11, 165-173.

Conrad, R. and Longman, D. J. A. Standard typewriter versus chord keyboard—an experimental comparison. *Ergonomics*, 1965, 8, 77-88.

Cornog, J. R., Hockman, J. F., and Craig, J. C. Address encoding—a study of the double-binary keyboard as a link in the machine-sorting of mail (paper 63 WA 338). American Society of Mechanical Engineers, 1963.

Creamer, L. R. and Trumbo, D. A. Multifinger tapping performance as a function of the direction of tapping movements. *Journal of Applied Psychology*, 1960, 44, 376-380.

Daniels, R. W. and Graf, C. P. The influence of keyset interlocks on operator performance (Report No. 12215-FR1). Honeywell Systems and Research Center, April 30, 1970.

Davidson, L. A pushbutton telephone for alphanumeric input—two extra buttons. *Datamation*, April 1966, 27-30.

Deininger, R. L. Human factors engineering studies of the design and use of pushbutton telephone sets. *Bell Systems Technical Journal*, 1960, 39, 995-1012.

Diehl, M. J. and Seibel, R. The relative importance of visual and auditory feedback in speed typewriting. *Journal of Applied Psychology*, 1962, 46, 365-369.

Dreyfuss, H. *Anthropometric data.* New York: Whitney Publications, 1959.

Droege, R. C. and Hill, M. Comparison of performance on manual and electric typewriters. *Journal of Applied Psychology*, 1961, 45, 268-270.

Dunn, A. G. Engineering of the typing keyboard from the human factors viewpoint. *Computers and Automation.* February 1971, 32-33.

Dvorak, A., Merrick, N. F., Dealey, W. L., and Ford, G. C. *Typewriting behavior: Psychology applied to teaching and learning typewriting.* New York: American Book Co., 1936.

Flanagan, C., Fivars, G., and Tuska, S. A. Predicting success in typing and keyboard operation. *Personnel and Guidance Journal*, 1959, 37, 353-357.

Fleishman, E. A. and Rich, S. Role of kinesthetic and spatial-visual abilities in perceptual-motor learning. *Journal of Experimental Psychology*, 1963, 66, 6-11.

Fox, J. G. and Stansfield, R. G. Digram keying times for typists. *Ergonomics*, 1964, 7, 317-320.

Galitz, W. O. CRT keyboard human factors evaluation. UNIVAC, Systems Application Engineering, Roseville DPD, March 1965.

Galitz, W. O. CRT keyboard human factors evaluation: Study II. UNIVAC, Systems Application Engineering, Roseville DPD, February 1966. (a)

Galitz, W. O. CRT keyboard human factors evaluation: Study III. UNIVAC Systems Application Engineering, Roseville DPD, June 8, 1966. (b)

Galli, E. J. The stenowriter system. IBM Research Report RC-308, July 1960.

Garrett, J. W. The adult human hand: Some anthropometric and biomechanical considerations. *Human Factors*, 1971, 13, 117-131.

Haaland, J. Anatomic and physiological consideration of the use of the thumb for manual control in space flight. Honeywell Memorandum, July 11, 1962.

Haaland, J., Wingert, J., and Olson, B. A. Force required to actuate switches, maximum finger-pushing force, and coefficient of friction of Mercury gloves. Honeywell Memorandum, February 23, 1963.

Harkins, W. H. Switch system for consoles. *Industrial Design*, 1965, 1-7.

Hillix, W. A. and Coburn, R. Human factors in keyset design (NEL/Report 1023). San Diego, California: U.S. Naval Electronics Laboratory, March 1961. (AD 622368)

Hirsch, R. S. Effects of standard versus alphabetical keyboard formats on typing performance. *Journal of Applied Psychology*, 1970, 54, 484-490.

Howell, C. and Kreidler, L. Information processing under contradictory instructional sets. *Journal of Experimental Psychology*, 1963, 65, 30-46.

Hufford, L. E. and Coburn, R. Operator performance on miniaturized decomalentry keysets (NEL/Report 1083). San Diego, California: U.S. Naval Electronics Laboratory, December 1961.

Kallage, R. Electronic keyboard design with *n*-key rollover. *Computer Design*, February 1972, 57-61.

Kinkead, R. D. and Gonzalez, Barbara K. Human factors design recommendations for touch-operated keyboards—final report (Document 12091-FR). Minneapolis: Honeywell Inc., March 1969.

Klemmer, E. T. Keyboard entry. *Applied Erogonomics*, 1971, 2, 2-6.

Klemmer, E. T. and Lockhead, G. R. Productivity and errors in two keying tasks: A field study. *Journal of Applied Psychology*, 1962, 46, 401-408. (a)

Klemmer, E. T. and Lockhead, G. R. Further data on card punch operator performance (Research note NC 39). Yorktown Heights, New York: IBM, Thomas J. Watson Research Center, 1962. (b)

Kramer, J. J. Human factors problems in the use of pushbutton telephones for data entry (VDE-VERLAG GooBtt). Proceedings of Fourth International Symposium on Human Factors in Telephony, Berlin, 1970, 241-258.

Kroemer, K. H. E. The effect of keyset layout on performance on typewriters. *International Zeitschrift angewandt Physiologie*, 1964, 20, 240-251.

Kroemer, K. H. E. Comparison of a keyboard of a normal typewriter with a "K-keyboard." *International Zeitschrift angewandt Physiologie*. 1965, 20, 453-464.

Lee, W. A. and Snodgrass, J. G. On the relation between numbering preferences and performance on a ten-button keyboard. Abstract in *American Psychologist*, 1958, 13, 425.

Leonard, J. A. and Carpenter, A. On the correlation between a serial-choice task and subsequent achievement of typewriting. *Ergonomics*, 1964, 7, 197-204.

Leonard, J. A. and Conrad, R. Maintenance of high accuracy without augmented feedback. *Nature*, 1963, 512-513.

Leonard, J. A. and Newman, R. C. On the acquisition and maintenance of high speed and high accuracy in a keyboard task. *Ergonomics*, 1965, 8, 281-304.

Lewin, M. H. Portable electronic keyboard for computer input by telephone. *IEEE Transactions on Electronic Computers*, 1967, EC-16, 332-334.

Lincoln, R. A. and Konz, S. A. Effect of switch configuration on the operation of a switch matrix. *Journal of Applied Psychology*, 1966, 50, 375-382.

Lohse, C. (Ed.) Proposed U.S.A. standard: General-purpose alphanumeric keyboard arrangement for information interchange. *Communications of the ACM*, 1968, 11, 126-129. (a)

Lohse, C. (Ed.) Proposed USASCSOCR dual-case keyboard arrangement. *Communications of the ACM*, 1968, 11, 130-131. (b)

Lundervold, A. J. S. Electromyographic investigations of position and manner of working in typewriting. *Acta Physiologica Scandinavica*, 1951, 24, (Supplementum 84).

McCauley, R. and Parkinson, R. The new popularity of the Dvorak simplified keyboard. *Computers and Automation*, February 1971, 31-32.

McCormick, E. J. *Human factors engineering*. (2nd ed.) New York: McGraw-Hill, 1964.

Martin, A. Predicting keyboard trainability. *Occupational Psychology*, 1956, 30, 216-227.

Michaels, S. E. QWERTY versus alphabetic keyboards as a function of typing skill. *Human Factors*, 1971, 13, 419-426.

Minor, F. J. Experimental comparison of two keyboard interlock systems. *Journal of Engineering Psychology*, 1964, 3, 9-15.

Minor, F. J. and Revesman, S. L. Evaluation of input devices for a data setting task. *Journal of Applied Psychology*, 1962, 46, 332-336.

Moneta, Katherine B. Optimum physical design of an alphabetical keyboard. Part I. Survey of the literature. Post Office Research Report No. 20412, Part 1, November 1960.

Newman, K. M., Coburn, R., and McMahan, R. C. Experimental study of binary coded keysets (NEL/Report 1137). San Diego, California: U.S. Naval Electronics Laboratory, September 1962. (AD 291796)

Paul, F., Sarlanis, K., and Buckley, E. P. A human factors comparison of two data entry keyboards. Paper presented at Institute of Electrical and Electronics Engineers Symposium on Human Factors in Electronics, Boston, May 1965.

Phillips, Sue Ann and Kincaid, J. P. The human engineering of the typewriter keyboard: A case in point. *Human Factors Society Bulletin*, July 1971, 14, 7-8.

Pollock, W. T. and Gildner, G. G. Study of computer manual input devices (ESD-TDR-63-545). USAF: Decision Sciences Laboratory, L. G. Hanscom Field, Bedford, Massachusetts, September 1963. (AD 419254)

Poulton, E. C. and Brown, C. H. Rate of comprehension of an existing teleprinter output and of possible alternatives. *Journal of Applied Psychology*, 1968, 52, 16-21.

Ratz, H. C. and Ritchie, D. K. Operator performance on a chord keyboard. *Journal of Applied Psychology*, 1961, 45, 303-308.

Remington, R. J. and Rogers, Marilyn. Keyboards: Human engineering design criteria (TR 29.0042). Research Triangle Park, North Carolina: IBM Systems Development Division, February 1969.

Rosberger, P. Double functions per unit space with binary keyboard. *Research/Development*, July 1965, 16, 35-38.

Scales, E. M. and Chapanis, A. The effect on performance of tilting the toll-operator's keyset. *Journal of Applied Psychology*, 1954, 38, 452-456.

Seibel, R. Performance on a five-finger chord keyboard. *Journal of Applied Psychology*, 1962, 46, 165-169.

Seibel, R. Data entry through chord, parallel entry devices. *Human Factors*, 1964, 6, 189-192.

Shaffer, L. H. and Hardwick, J. Errors and error detection in typing. *Quarterly Journal of Experimental Psychology*, 1969, 21, 209-213. (a)

Shaffer, L. H. and Hardwick, J. Reading and typing. *Quarterly Journal of Experimental Psychology*, 1969, 21, 381-383. (b)

Smith, S. L. and Goodwin, Nancy C. Alphabetic data entry via the touch-tone pad: A comment. *Human Factors*, 1971, 13, 189-190.

Smith, W. A. Jr. Accuracy of manual entries in data-collection devices, *Journal of Applied Psychology*, 1967, 51, 362-368.

Strong, E. P. A comparative experiment in simplified keyboard retraining and standard keyboard supplementary training. Washington, D.C.: General Services Administration, 1956.

United States military standards: Human engineering design criteria for aerospace systems and equipment. Part 1: Aerospace system ground equipment (MIL-STD-803 A-1). 1964.

Welford, A. T. The measurement of sensory-motor performance: Survey and reappraisal of twelve years' progress. *Ergonomics*, 1960, 3, 189-229.

West, L. J. Review of research in typewriting learning with recommendation for training (AFPTRC-TN-57-69). USAF: Air Force Personnel and Training Research Center, Lowry AFB, Colorado, June 1957. (AD 126400)

West, L. J. Vision and kinesthesis in the acquisition of typewriting skill. *Journal of Applied Psychology*, 1967, 51, 161-166.

West, L. J. Fatigue and performance variability among typists. *Journal of Applied Psychology*, 1969, 53, 80-86.

West, L. J. and Bolanovich, D. J. Evaluation of typewriting proficiency training: Preliminary test development. *Journal of Applied Psychology*, 1963, 47, 403-407.